Mixture of Receptive Fields Fusion for Accurate Rice Panicle Segmentation and Yield Prediction in UAV-Based Remote Sensing

Zibin Zhou, Xuan Wei, Huasheng Huang, Tao Guo, Jiajun Zhuang, Zhiping Tan, Jinchang Ren, *Senior Member, IEEE*, and Yu Tang, *Member, IEEE*

***Abstract*—Unmanned aerial vehicle (UAV)-based remote sensing enables high-throughput rice phenotyping, yet precise panicle segmentation remains challenging. Conventional plot-level vegetation indices (VIs) or panicle counting proxies inherently discard pixel-level structural information, rendering them incapable of organ-level morphological analysis. While ultrahigh-resolution (UHR) cameras now enable fine-grained segmentation, three interrelated bottlenecks persist: extreme scale heterogeneity across genotypes induces receptive field mismatch, dense canopy occlusion coupled with spectral similarity between withered leaves and panicles causes severe foreground–background confusion, and gigapixel-scale orthomosaics for thousands of microplots impose prohibitive computational burdens. To address these, we propose mixture of receptive fields fusion (MiRF), where dynamic preview convolution (DyPConv) adaptively selects optimal dilation rates to capture critical local contextual information across diverse genotypic panicle morphologies while compressing computation to only K active branches via hard path pruning; frequency-aware refinement (FAR) leverages Laplacian high-frequency components with learnable gating to explicitly magnify textural disparities between panicles and cluttered backgrounds of withered leaves; and efficient cross-scale feature aggregator (ECFA) employs globally shared bidirectional fusion to reduce computational burdens when processing gigapixel-scale orthomosaics while preserving multiscale context. Experiments on two public datasets achieve state-of-the-art intersection over union (IoU)-Panicles (+3.48% and +3.11%) and *F*1-scores (+2.83% and +2.31%) with significantly reduced complexity. Cross-season validation on 610 varieties spanning 2024–2025 achieves R^2 exceeding 0.826 for yield prediction, supporting MiRF's generalization across diverse genotypes. Edge deployment experiments demonstrate its superiority in real-time processing.**

***Index Terms*—Panicle segmentation, rice breeding, unmanned aerial vehicle (UAV)-based remote sensing, yield prediction.**

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Zibin Zhou and Xuan Wei are with the College of Mechanical and Electrical Engineering, Fujian Agriculture and Forestry University, Fuzhou 350003, China (e-mail: Zhou01425@163.com; xuanweixuan@126.com).

Huasheng Huang and Jinchang Ren are with the School of Artificial Intelligence, Guangdong Polytechnic Normal University, Guangzhou 510665, China (e-mail: huanghsheng@gpnu.edu.cn; jinchang.ren@ieee.org).

Tao Guo is with the College of Agriculture, South China Agricultural University, Guangzhou 510642, China (e-mail: guoguot@scau.edu.cn).

Jiajun Zhuang is with the Academy of Contemporary Agriculture Engineering Innovations, Zhongkai University of Agriculture and Engineering, Guangzhou 510225, China (e-mail: zhuangjiajun@zhku.edu.cn).

Zhiping Tan is with the Academy of Interdisciplinary Studies, Guangdong Polytechnic Normal University, Guangzhou 510665, China, and also with the College of Mechanical and Electrical Engineering, Fujian Agriculture and Forestry University, Fuzhou 350003, China (e-mail: tanzp@gpnu.edu.cn).

Yu Tang is with the Academy of Interdisciplinary Studies, Guangdong Polytechnic Normal University, Guangzhou 510665, China, and also with the School of Electronic and Electrical Engineering, Lingnan Normal University, Zhanjiang 524048, China (e-mail: yutang@gpnu.edu.cn).

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I. Introduction

RICE (*Oryza sativa L.*) sustains over half of the global population, yet accelerating breeding cycles to meet escalating food demands remains a formidable challenge [1], [2]. High-throughput phenotyping has emerged as a critical bottleneck in modern breeding pipelines. Among phenotypic indicators, panicle morphology directly determines grain number, filling rate, and ultimately yield potential, making its precise characterization indispensable for genotype selection.

Unmanned aerial vehicle (UAV) based remote sensing offers an unprecedented opportunity to capture panicle traits at scale [3], [4], [5]. However, translating raw imagery into biologically meaningful phenotypes remains fundamentally constrained by the physical imaging conditions inherent to agricultural scenes. During the reproductive stage, rice canopies exhibit complex light scattering among erect leaves, panicle rachises, and glumes. Meanwhile, the canopy architecture is strongly directional, meaning that observed reflectance significantly varies with viewing and illumination angles [4]. Conventional multispectral or red, green, and blue (RGB) sensors, while operationally efficient, typically operate at coarse spatial resolutions with ground sampling distances (GSDs) exceeding 1 cm. At this resolution, a single pixel often mixes multiple components, including panicles, leaves, and soil background, averaging their spectral signatures into a single composite response [6]. Consequently, organ-level morphological details are irreversibly lost during the imaging

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stage, and no downstream algorithm can recover the true panicle contours from such mixed signals.

This physical limitation has led existing studies to rely predominantly on plot-level indirect proxies, such as vegetation indices (VIs) derived from multispectral bands [6], [7], [8] or panicle counting methods based on object detection [9]. However, these approaches suffer from two interconnected deficiencies. First, VIs compress rich spectral information into simple ratios, discarding subtle spectral cues that could otherwise distinguish panicle glumes from withered leaves. Moreover, such indices typically assume a Lambertian surface, yet erect-leaf rice canopies form pronounced specular reflection under certain illumination-viewing geometries, severely violating this assumption [7]. Second, plot-level spatial aggregation averages all subpixel components, erasing pixel-level structural information. It also renders empirical relationships between VIs and yield across genotypes and growth environments, as variations in leaf area and leaf angle alter the light propagation path through the canopy and consequently perturb the regression surface [10].

To circumvent the spatial resolution bottleneck, 3-D point cloud reconstruction has been explored for individual panicle isolation [11], [12]. Such methods leverage multiview stereo or LiDAR sensing to recover geometric structure in three dimensions, enabling direct volumetric measurement of panicle architecture. However, LiDAR incurs prohibitive computational costs for large-area breeding fields [13], [14] and remains vulnerable to in-field illumination variability that disrupts photometric consistency and wind-induced motion artifacts that misalign sequential point sets. The operational overhead and environmental fragility of 3-D pipelines have limited their deployment in routine high-throughput breeding workflows.

Recently, the integration of large-format complementary metal–oxide–semiconductor (CMOS) sensors with low-altitude UAV platforms has enabled ultrahigh-resolution (UHR) imaging at submillimeter GSD. When spatial resolution is sufficient to resolve the edges and textures of individual panicles, the aforementioned mixed-pixel issue is substantially alleviated, making organ-level semantic segmentation feasible. Nevertheless, deploying such technology in practical breeding environments exposes three interrelated challenges to the mainstream deep learning architectures.

Extreme scale heterogeneity across genotypes induces receptive field mismatch. As shown in Fig. 1(b), breeding trials encompass varieties with substantial morphological divergence. Tall and high-tillering varieties with dense canopies produce larger and more dispersed panicles, while compact dwarf lines exhibit smaller and clustered panicle architectures. Under a fixed observation geometry and GSD, these structural differences translate into highly variable pixel-scale representations of panicles across the same orthomosaic. Standard encoder–decoder networks employ successive downsampling to enlarge receptive fields, irreversibly eroding fine-grained spatial information essential for delineating small targets. Multiscale fusion modules such as atrous spatial pyramid pooling (ASPP) or pyramid pooling modules (PPMs) attempt to mitigate this through parallel dilated convolutions or hierarchical pooling. However, ASPP incurs substantial computational overhead from redundant atrous operations [15], whereas PPM degrades pixel-level correlations through aggressive downsampling [16]. Neither approach dynamically adapts to spatially varying scales within a single image, resulting in suboptimal feature aggregation across heterogeneous panicle morphologies [17].

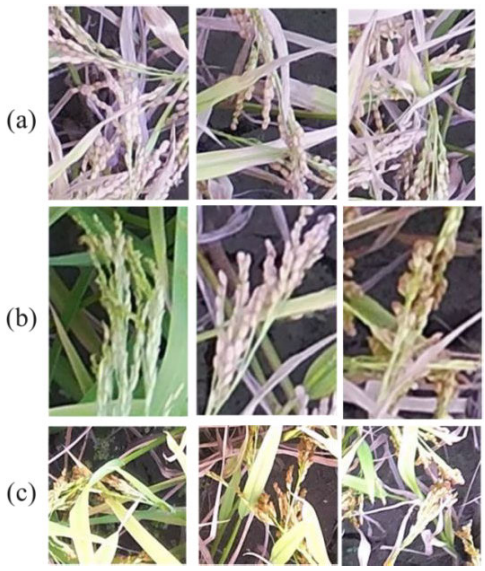


Fig. 1. Challenges of semantic segmentation in rice breeding. (a) Mutual occlusion and morphological similarity between panicles and leaves. (b) Interpanicle morphological diversity across varieties. (c) Uneven illumination.

Severe foreground–background confusion arises from radiative transfer complexity in dense canopies. As shown in Fig. 1(a)–(c), during reproductive maturity, withered leaves curl into elongated strips that share high spectral similarity in the visible domain. This phenomenon is exacerbated by the multiple scattering effects and anisotropic reflectance. In the lower canopy layers, scattered light from the soil background and senescent leaves intermingles with panicle radiance, reducing the spectral separability that would otherwise exist under uniform illumination. Existing frequency-domain methods for enhancement rely on static high-pass filtering or generic spatiofrequency fusion strategies [18], [19]. These designs inadequately adapt to the dynamic illumination conditions encountered during UAV sorties and to cultivar-specific texture patterns, leaving boundary regions ambiguous and phenotype extraction accuracy fundamentally limited.

Computational overhead limits real-world deployment. UHR imagery from 100-m pixel sensors at submillimeter GSD generates gigapixel-scale orthomosaics for even modest breeding fields. With thousands of microplots to be processed within narrow operational windows between optimal flight timing and harvest, the data throughput requirements far exceed the capacity of complex architectures with excessive floating-point operations (FLOPs). Conversely, lightweight designs that reduce latency often sacrifice representational capacity, particularly for the fine-grained textures that distinguish panicles from background clutter with high-resolution imagery. It is still unresolved to strike the balance between segmentation accuracy and processing efficiency for UAV-based high-throughput phenotyping.

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To address these challenges, we present mixture of receptive fields fusion (MiRF), a lightweight neural architecture specifically engineered for UHR rice panicle segmentation. Our contributions are threefold.

1) We propose dynamic preview convolution (DyPConv), a scale-adaptive mechanism that adaptively selects optimal dilation rates to capture critical local contextual information across diverse genotypic panicle morphologies. Unlike conventional dynamic convolution where all branches must be computed, DyPConv achieves trainable hard path pruning through a Union-of-TopK batch processing strategy, compressing computation to only K active branches while maintaining graphics processing unit (GPU) parallel efficiency, thereby enabling efficient multiscale feature aggregation at constant computational cost.
2) We introduce the frequency-aware refinement (FAR), a learnable-gated three-stream module combining spatial semantics, Laplacian high-frequency cues, and residual identity. The spatial stream captures global context; the Laplacian stream magnifies textural disparities between panicles and withered-leaf clutter to sharpen occlusion boundaries and suppress spectral confusion; and the identity stream prevents oversmoothing. Collectively, these streams alleviate boundary ambiguity and spectral confusion in dense canopies.
3) We establish an operational phenotyping pipeline integrating MiRF segmentation with lightweight yield regression. Comparative experiments on two benchmarks demonstrate state-of-the-art intersection over union (IoU) gains (+3.48% and +3.11%) over prevailing convolutional neural networks (CNNs) and Transformers with lower computational overhead, validating MiRF's superiority for heterogeneous panicle delineation. Deployed across 610 varieties spanning two seasons, the segmentation-based yield estimation achieves R^2 0.826, substantially outperforming conventional regression approaches.

The remaining article is organized as follows. Section II reviews related work on deep learning for crop phenotyping. Section III elaborates on the proposed MiRF architecture, detailing the reparameterized top-down (TD)/bottom-up (BU) pathways, DyPConv, and the three-stream FAR module. Section IV presents comprehensive experimental results, including ablation studies and comparisons with state-of-the-art methods, followed by the validation of our yield estimation framework on 610 rice varieties. Finally, Section VI concludes this study and outlines the future work.

II. Related Work

A. Semantic Segmentation of UAV Images

Semantic segmentation of UAV images faces significant challenges, particularly in dealing with small objects and boundary blurring. Recent research has focused on two main strategies to address these issues: receptive field expansion and boundary-aware feature refinement, as detailed in the following.

Receptive field expansion: The design of multibranch receptive field modules has evolved from static fusion to dynamic and lightweight architectures. The Inception module [20] introduced parallel convolutions with varying kernel sizes to capture diverse local structures. However, its reliance on fixed branches hampers adaptability for targets with scale variations, especially in small object segmentation where fine-grained details can be lost. To enhance context aggregation, the PPM [21] efficiently summarizes contextual cues, while ASPP [22] utilizes parallel dilated convolutions to expand receptive fields without losing spatial resolution. Nevertheless, these approaches can lead to boundary blurring or high computational costs. Innovations such as InceptionNeXt [23] decompose large kernels into lightweight branches, improving processing speed. Selective kernel attention [24] dynamically reweights parallel branches, enhancing adaptability to scale changes, while multiscale dilated attention [25] integrates dilated attention with multihead mechanisms, facilitating better multiscale semantic fusion.

Boundary-aware feature refinement: To mitigate boundary blurring, researchers use edge modeling or frequency enhancement techniques. Generative methods such as EdgeConnect [18] reconstruct edge maps to improve structural coherence but depend on static Canny priors. FreqFusion [26] combines adaptive low-/high-pass filtering with a dynamic offset generator to sharpen boundaries, while frequency-dynamic attention modulation (FDAM) [27] adapts attention mechanisms in the frequency domain to recover high-frequency components often suppressed by deep networks. Other literature exploits the collaboration of edge modeling and frequency enhancement, building joint implicit-explicit modeling strategies for boundary refinement. Hybrid modules, such as FusedEnhanceBlock [28] and SFHformer [29], combine Fourier representations and spatial convolutions for precise boundary refinement. This design bridges structural understanding of global context (frequency cues) and local continuity (spatial features), yet it remains limited by static weighting or task-generic designs.

In contrast to these methods, this work proposes DyPConv and the FAR module to improve small object segmentation and mitigate boundary blurring in rice UAV images' semantic segmentation.

B. Rice Yield Prediction

Current rice yield prediction is dominated by two mainstream approaches: satellite remote sensing and UAV multispectral analysis. However, both approaches suffer from critical limitations in rice phenotyping at microplot scale, causing scale mismatch and poor model interpretability.

Satellite remote sensing-based methods have realized field-level yield prediction through multisource data fusion, with typical models achieving R^2 values ranging from 0.55 to 0.99 [30], [31]. Nevertheless, constrained by meter-level spatial resolution, these methods are inherently unable to capture the fine-grained details in decimeter-scale breeding plots. This scale mismatch makes them incompatible with the precise evaluation of rice yield, as they fail to reflect the subtle phenotypic variations within small plots [30], [32].

UAV multispectral analysis has emerged as a focus for small-scale yield prediction. Su et al. [33] utilized multitem-

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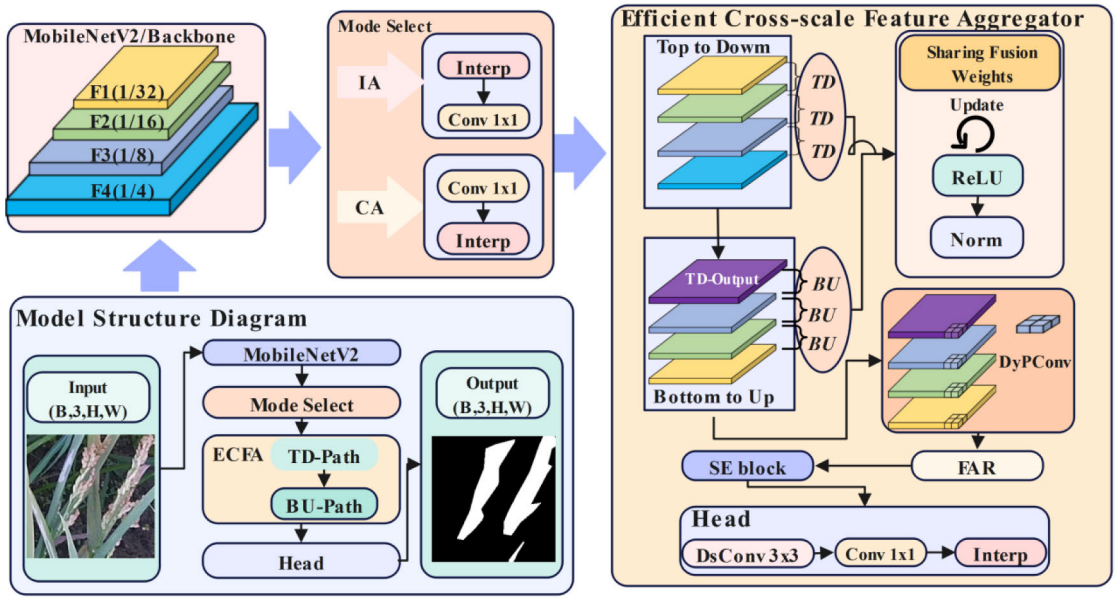


Fig. 2. Overall workflow of our proposed MiRF model.

poral UAV-derived VIs and endmember abundance with the XGBoost model, achieving reliable prediction performance. Similar UAV-based studies have confirmed that fusing VIs of key growth stages can improve prediction accuracy. Despite better adaptability to small scales than satellite remote sensing, UAV multispectral imaging is still limited by the spatial resolution for fine-grained crop phenotyping. More importantly, machine learning models, such as XGBoost and deep learning models, suffer from poor interpretability [33], [34], failing to establish correlations between the input features (VIs and images) and rice yield formation mechanisms, which limits their support for breeding decision-making.

In contrast to these existing methods, we target the scale and interpretability bottlenecks in rice phenotyping at microplot scale. This study aims to develop a prediction framework that can integrate UAV UHR image segmentation and dynamic feature fusion, which fills the research gap in current studies and provides more technical support for rice breeding in remote sensing.

III. Methodology

This section elaborates on the proposed methodology. First, we present the overall architecture of the MiRF model, as shown in Fig. 2, which comprises a backbone, efficient cross-scale feature aggregator (ECFA), and a simple head, as detailed in the following.

A. Dynamic Preview Convolution

Given an input feature map $x \in R^{B\times C\times H\times W}$, DyPConv employs a set of depthwise branches with different dilation rates $r = \{r_1, \ldots, r_R\}$ to explore multirate receptive fields and then computes only the selection of the K highest-ranked candidates (Top-K) promising branches end-to-end. As such, it decouples a computationally inexpensive, shared preview-and-selection stage from an expensive pointwise projection stage, thereby adaptively reallocating computation to the most informative scales on the input signals, as shown in Fig. 3.

For each dilation rate $r \in R$, a 3×3 depthwise convolution with dilation r first produces a preview feature

$$z_r = Dr(x) \in R^{B\times C\times H\times W} \tag{1}$$

where r indicates the dilation rate and Dr denotes the depthwise convolution operation with the dilation rate r.

Global average pooling then compresses each preview into a channel descriptor

$$v_r = GAP(z_r) \in R^{B\times C} \tag{2}$$

where GAP represents the global average pooling operation that reduces the spatial dimension from $H \times W$ to 1×1, preserving batch and channel dimensions. A lightweight function f_θ maps each descriptor to a scalar

$$a_r = f_\theta(v_r) = w_2^\top \sigma(W_1 v_r + b_1) + b_2 \tag{3}$$

where $W_1 \in R^{d\times C}$ and $b_1 \in R^d$ denote the weight matrix and bias vector of the first linear layer, σ represents the activation function, $w_2 \in R^d$ and $b_2 \in R$ correspond to the weight vector and bias scalar of the second linear layer, and d indicates the dimension of hidden layers.

Next, a temperature-controlled softmax function is employed to normalize the preview weights that estimate the utility of each rate r for the current input

$$p_r = \frac{exp(a_r/\tau)}{\sum_{j=1}^{R} exp(a_j/\tau)}, \tau > 0 \tag{4}$$

where τ represents the temperature hyperparameter that controls the sharpness of the weight distribution, R denotes the total number of branches, and p_r indicates the selection probability weight for branch r. To achieve sparse computation

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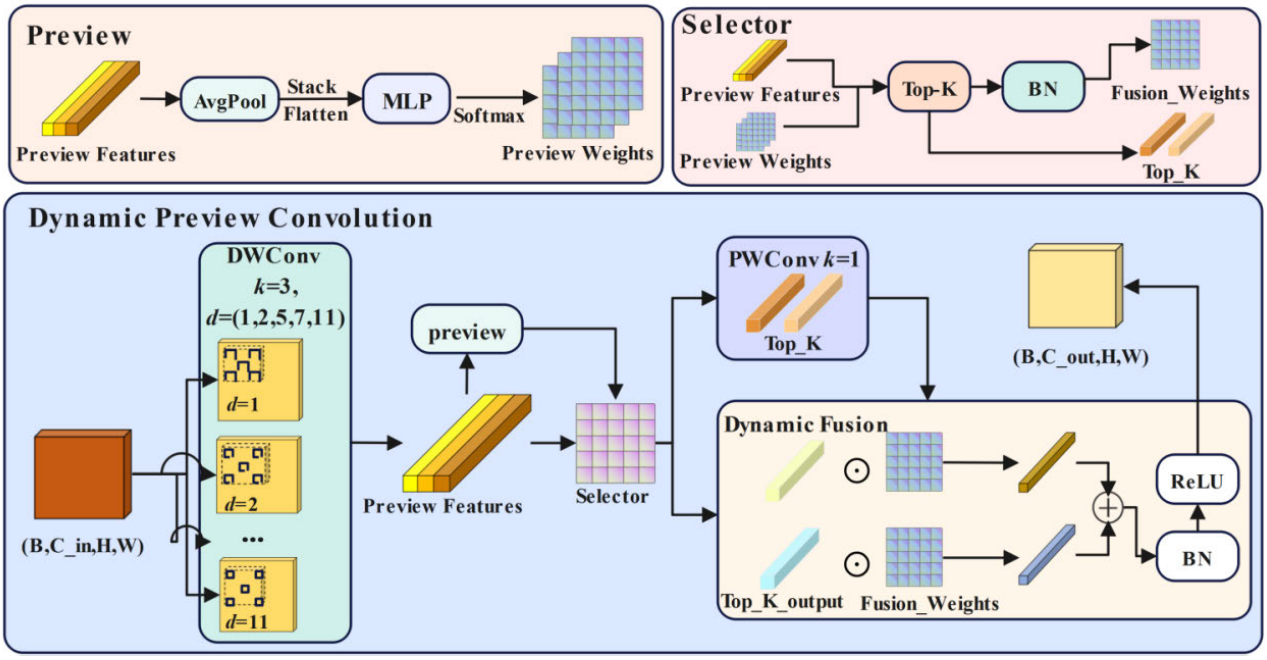


Fig. 3. Architecture of DyPConv.

while maintaining computational efficiency in batch processing, DyPConv selects the union of Top-K indices across all samples in a minibatch

$$S = TopK\left(p^{(b)}\right), \quad K' = |S| \le min\left(R, K \cdot B\right); m_r = 1 | r \in S \tag{5}$$

where TopK selects the K branches with the highest probability weights, K' represents the actual number of selected branches, and the union operation ensures GPU-friendly batch processing. Here, S denotes the set of unique active branch indices collected from the Top-K selections of all samples in the batch. Nonselected branches are set to zero using a binary mask $m_r = 1$ for $r \in S$, and the remaining weights are renormalized for each sample to preserve differentiability while enforcing strict computational sparsity

$$\alpha_r^{(b)} = \frac{m_r p_r^{(b)}}{\sum_{j=1}^{R} m_j p_j^{(b)} + \varepsilon}, \quad \sum_{r=1}^{R} \alpha_r^{(b)} = 1 \tag{6}$$

where $m_r \in \{0, 1\}$ represents the binary selection mask, ε is a small constant for numerical stability, and α_r indicates the renormalized fusion weight used for each selected branch.

The selected branches in S are subsequently processed by a 1×1 pointwise convolution

$$y_r = P_r\left(z_r\right) \in \mathrm{R}^{B \times C_{out} \times H \times W}, \quad y^{(b)} = \sum_{r \in S} \alpha_r^{(b)} y_r \tag{7}$$

where $r \in S$, P_r denotes the pointwise convolution operation, C_{out} represents the number of output channels, and y_r corresponds to the transformed feature map from branch r.

The outputs are then fused using the normalized weights to produce the final representation

$$y = \sum_{r \in S} \alpha_r y_r \tag{8}$$

where the summation combines selected branch outputs weighted by their respective importance scores α_r.

A standard postfusion block finalizes the transformation for stability and compatibility

$$\hat{y} = ReLU\left(BN\left(y\right)\right). \tag{9}$$

A practical memory–computation tradeoff is implemented through a weight reuse mechanism: when reuse-preview is enabled, the depthwise previews z_r are cached and directly fed into the pointwise layers; otherwise, previews are discarded to reduce memory footprint and recomputed only for the selected branches.

B. Frequency-Aware Refinement

The FAR module enhances boundary perception and high-frequency detail with a triple-stream architecture, as shown in Fig. 4. It processes original features, high-frequency components, and multiscale edge information in parallel, thereby explicitly counteracting the attenuation of high-frequency signals commonly observed in semantic segmentation while preserving computational efficiency through channelwise attention and lightweight convolutions. Given an input feature map $\mathrm{x} \in \mathrm{R}^{\mathrm{B} \times \mathrm{C} \times \mathrm{H} \times \mathrm{W}}$, FAR constructs two complementary representations alongside the original input. The high-frequency stream employs a fixed Laplacian high-pass kernel *Hhp* applied to the channelwise average $\mu ch(x)$, with the response replicated across channels and scaled for numerical stability

$$h_f = 0.1 \cdot repeat_C\left(\mathrm{Conv2D}\left(\mu_{ch}\left(x\right), H_{hp}, pad = 1\right)\right) \tag{10}$$

where $\mathrm{x} \in \mathrm{R}^{\mathrm{C} \times \mathrm{H} \times \mathrm{W}}$ denotes the input feature map with C channels. μ_{ch} represents the channelwise global average operation, compressing the input into a single-channel intensity map. H_{hp} refers to the fixed Laplacian high-pass kernel used to extract spatial boundary cues. To align the dimensions for subsequent fusion, $repeat_C$ broadcasts the single-channel edge map C times along the channel dimension. Finally, the scalar

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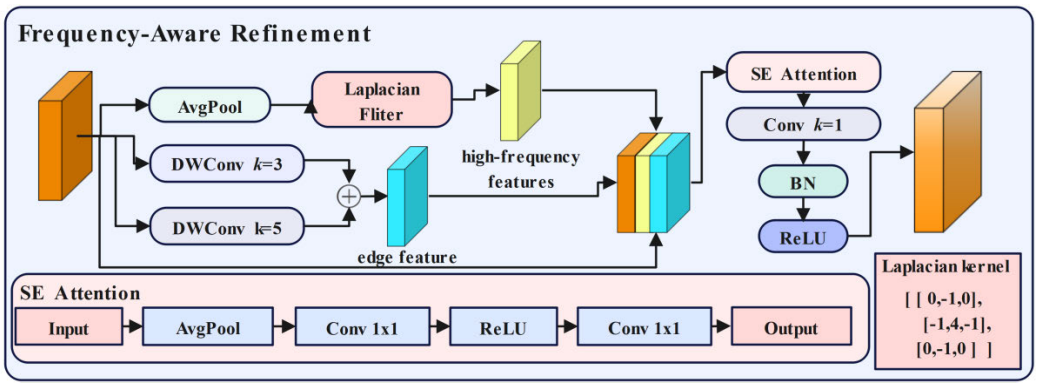


Fig. 4. Architecture of the FAR module.

0.1 serves as a weighting coefficient to stabilize the magnitude of high-frequency components.

In parallel, the edge stream aggregates multiscale boundary cues via depthwise convolutions at distinct receptive fields

$$edge = 0.5 \cdot \left(DWConv_{3\times3}(x) + DWConv_{5\times5}(x)\right) \tag{11}$$

where DWConv denotes the depthwise convolution operation with a kernel size of $k \times k$. Two distinct kernel sizes, 3×3 and 5×5, are employed in parallel to capture spatial structural details at different scale-fine-grained textures and coarser semantic boundaries, respectively. The resulting feature maps are summed and scaled by a factor of 0.5, which serves as a normalization term to maintain the magnitude consistency of the fused edge features.

The three streams, original features x, high-frequency components h_f, and edge features $edge$, are concatenated along the channel dimension

$$stacked = concat\left(x, \mathrm{h}_f, edge\right) \in \mathrm{R}^{B\times3C}. \tag{12}$$

A frequency-aware channel-attention mechanism then produces adaptive weights to emphasize informative responses across streams: let $z = GAP(stacked) \in \mathrm{R}^{B\times3C}$, and the attention vector $a = \sigma(W_2\phi(W_1 z)) \in \mathrm{R}^{B\times3C}$ is reshaped into $\{a_0, a_1, a_2\}$, with $a_i \in \mathrm{R}^{B\times C\times1\times1}$ and applied channelwise to each stream yielding $w_i = stream_i \odot a_i, i \in \{0, 1, 2\}$. The weighted streams are concatenated and fused through a pointwise convolution followed by batch normalization and ReLU, yielding the refined output as

$$\hat{y} = \delta\left(BN\left(W_f \cdot concat\left(w_0, w_1, w_2\right)\right)\right) \tag{13}$$

where w_i is the weighted feature of the ith stream, $W_f \in \mathrm{R}^{C\times3C}$ denotes the 1×1 fusion kernel, and δ is the ReLU function.

In contrast to conventional refinement modules that rely exclusively on spatial attention or channel attention, FAR explicitly integrates frequency-domain processing via the high-pass branch while maintaining spatial coherence through multiscale edge detection. The triple-stream architecture with adaptive weighting preserves and enhances high-frequency detail and boundary information with minimal computational costs, rendering the module particularly suitable for remote sensing imagery (RSI) semantic segmentation that demands precise localization and detail preservation.

C. Efficient Cross-Scale Feature Aggregator

To realize stable and efficient cross-scale aggregation under tight computational budgets, we propose ECFA, a concise pipeline of lateral channel alignment, a bidirectional progressive fusion, and an intralevel lightweight refinement given multiscale backbone features $\{F_1, F_2, F_3\}$ from high to low resolutions; we first integrate all scales via convolution with normalization and activation, as shown in the following:

$$T_\mathrm{l} = \delta\left(BN\left(W_{lat}^{(\mathrm{l})} * F_\mathrm{l}\right)\right), \quad W_{lat}^{(\mathrm{l})} \in \mathrm{R}^{C_d\times C_\mathrm{l}\times1\times1} \tag{14}$$

where δ denotes the ReLU activation function, BN represents batch normalization, and $*$ is the convolution operation. $W_{lat}^{(\mathrm{l})}$ is the 1×1 convolutional kernel that projects the features from the backbone channel dimension C_l to the unified decoding dimension C_d. Consequently, F_l is the input feature map, and T_l is the transformed feature with aligned channels.

After lateral alignment, ECFA performs TD and BU progressive fusion using shared scalar parameters across all pyramid levels. This global parameter sharing avoids levelwise parameter inflation while improving optimization robustness. In the TD path, context from the lowest resolution is progressively propagated to higher resolutions. Let $\tilde{T}_\mathrm{l}$ denote TD-updated features, $\alpha = \sigma(\theta_{TD}) \in (0, 1)$ is the scale-level fusion weight, and Up refers to bilinear upsampling

$$\tilde{T}_3 = \mathrm{T}_3, \quad \tilde{T}_\mathrm{l} = \alpha\mathrm{T}_\mathrm{l} + (1-\alpha)\,\mathrm{Up}\left(\tilde{T}_{\mathrm{l}+1}\right), \quad l = 1, 2. \tag{15}$$

The BU path complements high-resolution detail with stronger semantics and global information. Let $\hat{B}_\mathrm{l}$ denote BU-updated features, $\beta = \sigma(\theta_{BU}) \in (0, 1)$ represent the scale-level fusion weight, and Dℓ refer to a downsampling operator implemented by depthwise separable convolution, and then,

$$\hat{B}_\mathrm{l} = \tilde{T}_\mathrm{l}, \hat{B}_{\mathrm{l}+1} = \beta\tilde{T}_{\mathrm{l}+1} + (1-\beta)\,\mathrm{D}_\mathrm{l}\left(\hat{B}_\mathrm{l}\right), \quad l = 1, 2$$
$$\mathrm{D}_\mathrm{l}(\mathrm{x}) = \delta\left(\mathrm{BN}\left(\mathrm{W}_{\mathrm{pw}}^{(\mathrm{l})} * \left(\mathrm{W}_{\mathrm{dw}}^{(\mathrm{l})} * x\right)\right)\right) \tag{16}$$

where $*$ denotes the convolution. The scalars α and β are globally shared across pyramid levels during both training and inference, exactly corresponding to the implemented scale-level fusion weights.

On top of the bidirectionally fused features $\{B^\ell\}$, ECFA applies an intralevel refinement sequence, including DyPConv, FAR, and lightweight squeeze-and-excitation (SE) attention, to simultaneously enhance multirate context and boundary

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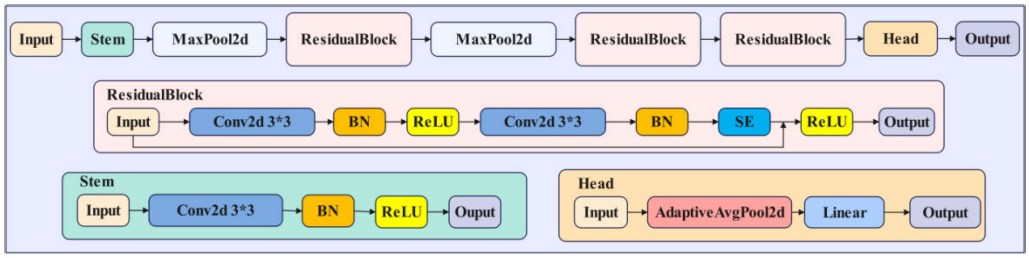


Fig. 5. Architecture of the rice yield prediction model.

separability. Denote the dynamic preview multirate module, the FAR, and the lightweight SE as M_l, R_l, and S_l, respectively. For each level $\ell \in \{1, 2, 3\}$, we have

$$\tilde{P}_l = M_l\left(\hat{B}_l\right), \quad \overline{P_l} = R_l\left(\tilde{P}_l\right), \; P_l = S_l\left(\overline{P_l}\right) \tag{17}$$

where M_l gates and fuses a small set of dilation branches selected by preview signals, R_l performs channelwise weighting and 1×1 convolution across the original, high-frequency, and edge streams to sharpen boundaries, and S_l recalibrates channels. The resulting $\{P_1, P_2, P_3\}$ form the pyramid outputs, which are employed for the subsequent decoding.

To balance training stability and inference efficiency, ECFA adopts an "interpolate then aggregate (IA) strategy for training, convolve then aggregate (CA) strategy for inference," which is theoretically equivalent due to the linear nature of both interpolation and convolution operations. The order of these operations can be exchanged without affecting the final output, as the commutative property holds when combining linear transformations. During training, we first upsample each level and then apply a bias-free 1×1 convolution to the feature after concatenation

$$X_l = BL(P_l; s_l), \quad X_o = Conv_{1\times1}([X_1, X_2, X_3]; G_{IA}) \tag{18}$$

where BL denotes bilinear interpolation and G_{IA} represents the 1×1 convolutional kernel for training-time feature aggregation.

During inference, we apply a levelwise 1×1 convolution before upsampling and summation

$$Y_l = Conv_{1\times1}\left(P_l; G_{CA}^{(l)}\right), X_o = \sum_l BL(Y_l; s_l). \tag{19}$$

As bilinear interpolation can be decomposed into scalar multiplication and addition, and given that a bias-free 1×1 convolution is a linear map, it can be concluded that

$$\begin{aligned} &\mathrm{Conv}_{1\times1}\left(\left[\mathrm{BL}(\mathrm{P}_1), \mathrm{BL}(\mathrm{P}_2), \mathrm{BL}(\mathrm{P}_3)\right]; \mathrm{G}_{\mathrm{IA}}\right) \\ &\quad = \sum_l BL\left(\mathrm{Conv}_{1\times1}\left(\mathrm{P}_l; \mathrm{G}_{\mathrm{CA}}^{(l)}\right)\right). \end{aligned} \tag{20}$$

If the IA kernel is the channelwise concatenation of CA kernels, it can be rewritten as

$$G_{IA} = \left[G_{CA}^{(1)}, G_{CA}^{(2)}, G_{CA}^{(3)}\right]. \tag{21}$$

While the model is trained using IA for stability and alignment, we deploy with the equivalent CA reparameterization to minimize FLOPs and latency. This zero-cost conversion yields identical outputs without fine-tuning. Furthermore, the TD/BU fusion weights persist as globally shared scalars across both phases, ensuring consistent cross-scale information.

D. MiRF-Based Yield Prediction Model

To establish a lightweight regression framework that maps MiRF-generated panicle features to grain yield, we design a streamlined residual network, as illustrated in Fig. 5. The architecture utilizes residual connections to ensure stable gradient propagation across diverse genetic backgrounds. SE modules are integrated to adaptively recalibrate channelwise responses to capture yield-related phenotypic cues, such as grain plumpness and maturity colors with minimal computational overhead.

The network takes segmented panicle regions of interest (ROIs), generated by applying MiRF binary masks to original UAV imagery, as input, effectively eliminating background interference and forcing the model to learn exclusively from yield-correlated morphological features. This design prioritizes deployment efficiency to enable real-time inference on UAV edge devices, serving as a validation tool for MiRF's segmentation utility in high-throughput yield estimation.

IV. Experiments

A. Datasets' Description

To comprehensively evaluate the proposed method, we utilized a total of four datasets. These are categorized into two groups based on their experimental purpose: panicle segmentation and rice yield prediction.

1) Datasets for Panicle Segmentation: Two RGB-image datasets characterized by varying spatial resolutions and a broad spectrum of rice genotypes (which induce substantial phenotypic differences in panicle shape, size, and color) were employed for training the MiRF model and benchmarking its segmentation performance.

a) UAV_PANICLE [35]: The UAV_Panicle dataset was constructed using high-resolution imagery collected during the 2024 early season from a standardized experimental field in Guangzhou, China, utilizing a Da-Jiang Innovations (DJI) M300 RTK UAV equipped with a 100-m pixel RGB sensor. The data acquisition focused on 12 distinct rice varieties during the maturity stage at an altitude of 12 m. To facilitate efficient model training for pixel-level semantic segmentation, the generated orthomosaic was cropped into 256×256 patches. A rigorous dataset partition strategy was employed to ensure generalization: eight varieties were randomly selected for training (8640 images) and validation (2160 images) with an 8:2 split, while the remaining four varieties constituted an independent test set (2048 images) to evaluate performance on unseen varieties.

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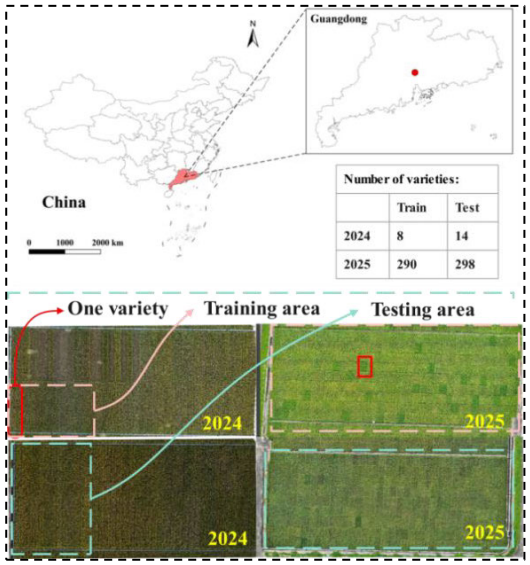


Fig. 6. Geographic location and experimental layout for the 2024 and 2025 cropping seasons.

b) Rice_seg_panicles [36]: To benchmark model generalization across diverse cultivars, we utilized RiceSEG, a public dataset comprising 565 UAV and ground-based images (512 × 512 pixels). This dataset features exceptional genetic and environmental diversity, covering over 6000 genotypes (Indica and Japonica) from five countries under complex illumination conditions. While the original dataset contains six semantic categories, we exclusively extracted panicle annotations and merged the remaining classes into the background to construct a binary segmentation dataset. The processed data were randomly partitioned into training and testing sets at the cultivar level with an 8:2 ratio, ensuring that no identical genotype appeared in both splits and, thus, preventing variety-level data leakage.

2) Datasets for Yield Prediction: To validate the robustness and generalization of our yield prediction model, we constructed two independent datasets spanning different cropping seasons under distinct experimental designs (see Fig. 6). The segmentation model was trained and evaluated on UAV_Panicle and Rice_SEG_Panicles, whereas yield prediction was conducted on independent 2024 late-season and 2025 early-season field datasets. Thus, no segmentation-training image patches were reused for yield regression. For both datasets, the ground truth of rice yield was obtained through manual harvesting, threshing, drying, and weighing.

a) 2024 late season: This experiment focused on 22 rice varieties planted in large plots (8 columns × 60 rows). Yield measurement involved each variety, with the ground truth recorded in kilograms (kg). To evaluate model performance on unseen cultivars, the dataset was partitioned into eight varieties for training and 14 varieties for testing.

b) 2025 early season: This season significantly expanded the genetic diversity to cover 588 varieties, planted in smaller breeding plots (8 columns × 14 rows). To avoid the edge effect, yield measurement was strictly limited to the central region (4 columns × 8 rows), with results recorded in grams (g). The dataset was split into 290 varieties for training and 298 varieties for testing, providing a rigorous benchmark for large-scale phenotypic analysis.

B. Implementation Details

In this study, we utilized an NVIDIA GeForce RTX 3090 GPU (24 GB) for model training and evaluation, adopting the AdamW optimizer with a base learning rate of 1e-3 and a weight decay of 0.01. To enhance generalization, all datasets were augmented with random scaling, random flipping, and random cropping, and all models were trained for 200 epochs. For all compared methods, backbone networks were initialized with ImageNet-1K pretrained weights to ensure a fair comparison.

Two types of objective metrics were used to intuitively compare the overall performance of the proposed segmentation model and other models. The first category is used to evaluate the accuracy of the segmentation results, including the overall accuracy (OA), $F1$-score ($F1$), and the mean IoU. Since rice panicles are closely correlated to yield production, both the $F1$ and IoU metrics are computed on the panicle class. The second category evaluates the complexity of models, including the FLOP count (FLOPs) and the number of model parameters (M).

For the evaluation of the yield prediction model, we employed three standard regression metrics: root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2). Specifically, RMSE and MAE were utilized to quantify the magnitude of deviation between the predicted yield and the actual ground truth, serving as direct indicators of prediction error where lower values represent better precision. Furthermore, R^2 was used to assess the goodness of fit, reflecting the proportion of variance in the yield data that are explained by the model, with a value closer to one indicating superior predictive performance.

C. Comparative Experimental Results

1) Quantitative and Visual Comparison of Segmented Panicle: As shown in Table I, the notable superiority of our method can be observed from the quantitative comparison with other counterparts. With ResNet18 as the shared backbone, conventional CNN-based models, such as UNet++, PSPNet, and DeepLabV3+, achieve around 0.62 in IoU, while enhanced convolutional variants, including ABCNet, UNetFormer, and SegFormer, slightly improve segmentation performance to about 0.63 by incorporating feature fusion and attention mechanisms. In contrast, Transformer-based models perform less favorably, with achieved IoU scores of 0.5748 and 0.5966, respectively. This inferior performance stems from a mismatch between global self-attention and the nature of panicle segmentation. Precise panicle delineation relies on local textures and boundary cues to discriminate glumes from morphologically similar distractors, whereas transformers aggregate long-range contexts that oversmooth fine-grained details and introduce noise from distant, structurally irrelevant regions. Under limited training samples, compact CNNs naturally prioritize local receptive fields and prove more effective for this fine-grained organ-level task.

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TABLE I
COMPARISON RESULTS ON UAV_PANICLE AND RICE_SEG_PANICLES DATASETS

		UAV_Panicle					Rice_SEG_Panicles					Params
	backbone	F1	IoU	OA	GFLOPs	FPS	F1	IoU	OA	GFLOPs	FPS	
UNet++[37]	ResNet18	0.7221	0.6288	0.9055	12.634	502	0.8770	0.7809	0.9560	50.51	110	26.138
DeepLabV3+[38]	ResNet18	0.7725	0.6294	0.9115	4.578	1665	0.8422	0.7274	0.9441	18.31	515	12.329
ABCNet[39]	ResNet18	0.7711	0.6275	0.915	3.847	2183	0.8612	0.7562	0.95	15.388	605	13.518
UNetFormer[40]	ResNet18	0.7706	0.6268	0.907	2.937	1978	0.8525	0.7429	0.9479	11.748	518	11.724
PSPNet[21]	ResNet18	0.7709	0.6272	0.911	1.442	**4294**	0.8587	0.7524	0.95	5.767	**1154**	11.33
MaSSFormer[41]	ResNet18	0.7424	0.5973	0.8990	26.12	326	0.8383	0.7351	0.9326	104.49	7	14.22
SegFormer[42]	ResNet18	0.7751	0.6327	0.9142	4.136	1513	0.866	0.7637	0.9522	16.544	348	11.768
TransUNet[43]	ViT-tiny	0.73	0.5748	0.897	3.75	383	0.8127	0.6845	0.9328	14.999	82	8.527
DPT[44]	ViT-tiny	0.7473	0.5966	0.9046	2.45	1750	0.8598	0.7541	0.9499	9.783	309	12.695
CMTFNet[45]	ResNet50	0.7708	0.6271	0.9062	8.862	429	0.8713	0.772	0.9542	34.238	122	30.069
CTCFNet[46]	/	0.7665	0.6214	0.9119	111.942	154	0.8634	0.7596	0.9526	450.536	30	73.849
GCNet[47]	/	0.7499	0.6068	0.9030	3.21	1186	0.8266	0.7199	0.9306	12.78	346	21.19
MiRF(ours)	ResNet18	**0.8034**	**0.6675**	**0.9249**	4.232	399	**0.8944**	**0.8031**	**0.9594**	16.926	240	12.09
MiRF(ours)	MobileNetV2	0.7859	0.6527	0.9224	**1.048**	2350	0.8780	0.7825	0.9559	**4.601**	590	**2.82**

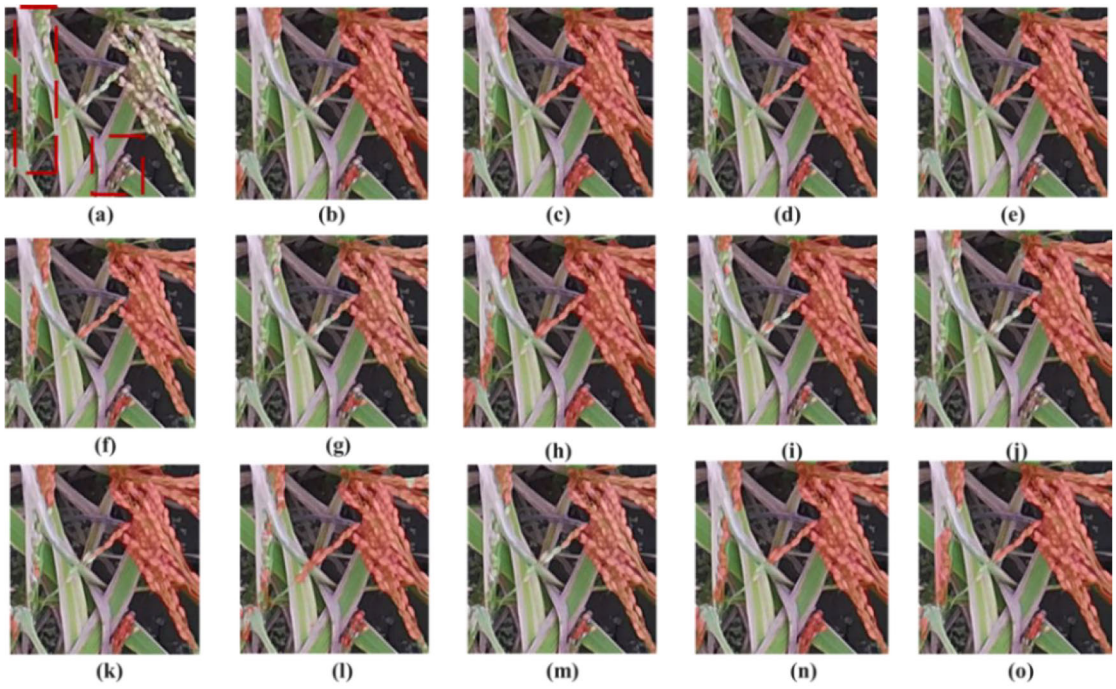


Fig. 7. Visualized comparison with the mainstream model of semantic segmentation on the UAV_Panicle dataset. (a) Original. (b) UNet++. (c) DeepLabV3 +. (d) ABCnet. (e) UNetFormer. (f) PSPNet. (g) MaSSFormer. (h) SegFormer. (i) TransUNet. (j) DPT. (k) CMTFNet. (l) CTCFNet. (m) GCNet. (n) MiRF_ResNet18. (o) MiRF_MobileNetV2.

The proposed MiRF balances local and global representations through multiscale feature fusion and lightweight semantic enhancement. It achieves the best results on this dataset in terms of improved IoU (0.6675, +3.48%) and $F1$ (0.8034, +2.83%) while maintaining moderate computational complexity. The lightweight version using MobileNetV2 attains IoU (0.6527, +2.0%) with only 2.82 M parameters, demonstrating strong adaptability and efficiency for natural field conditions.

As illustrated in Fig. 7, rice panicles are often fragmented into small scattered parts when occluded by yellow leaves with similar morphology and color, whereas MiRF

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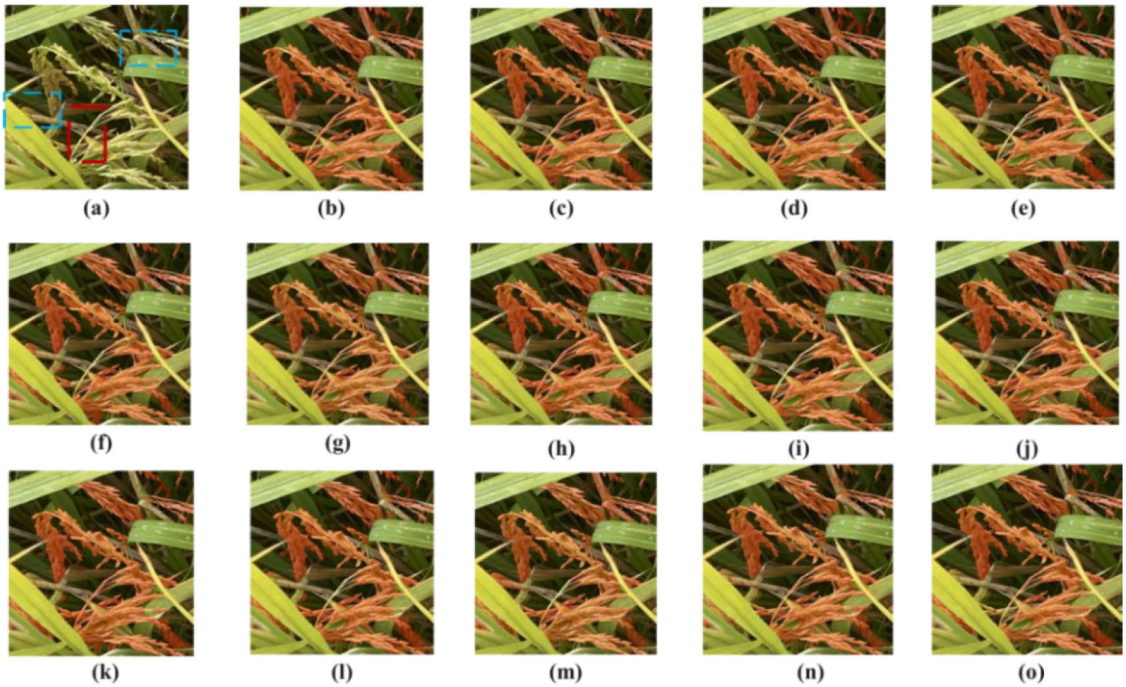


Fig. 8. Visualized comparison with the mainstream model of semantic segmentation on the RiceSEG dataset. (a) Original. (b) UNet++. (c) DeepLabV3 +. (d) ABCnet. (e) UNetFormer. (f) PSPNet. (g) MaSSFormer. (h) SegFormer. (i) TransUNet. (j) DPT. (k) CMTFNet. (l) CTCFNet. (m) GCNet. (n) MiRF_ResNet18; (o) MiRF_MobileNetV2.

demonstrates robust capability in accurately identifying and precisely segmenting these challenging regions. Furthermore, it can be seen from Table I that the proposed MiRF method is embedded in different networks and achieves the state-of-the-art performance, revealing that this method is adaptive to different model architectures.

2) Numerical and Visual Results on RiceSEG: As shown in Table I, segmentation results vary notably across different model architectures. With ResNet18 as the backbone, conventional CNN-based models, such as UNet++, PSPNet, and DeepLabV3+, achieve IoU scores ranging from 0.73–0.77, indicating stable and reliable local feature learning. Hybrid attention-enhanced architectures, such as ABCNet and UNetFormer, produce subtle improvements in segmentation accuracy through attention-based feature refinement, achieving IoU scores of 0.7562 and 0.7429, respectively. Transformer-based architectures, such as TransUNet and dense prediction transformer (DPT), benefit from global context modeling. Their IoU scores indicate that the long-range dependency modeling inherent to such architectures may not adapt well to the limited training set. Moreover, CTCFNet attains an IoU of 0.7596, but its extremely high complexity (over 73.8 M parameters and 450.5 GFLOPs) makes it impractical for UAV-based rice phenotyping.

In contrast, the proposed MiRF achieves the best overall performance on RiceSEG, with improved IoU (0.8031, +3.11%), $F1$ (0.8944, +2.31%), and OA (0.9594), while maintaining moderate complexity (12.09 M params and 16.93 GFLOPs). Its lightweight configuration using MobileNetV2 sustains competitive accuracy (IoU: 0.7825, $F1$: 0.8780) with a dramatic reduction in computational cost, i.e., 2.71 M params and 4.19 GFLOPs. As illustrated in Fig. 8, the occlusion between leaves and panicles makes it difficult to segment small-sized fragments of leaves, which are misclassified by most counterparts. In comparison, MiRF presents notable improvements in addressing these challenges, highlighting its advantage in establishing robust local context and segmenting small objects. These results reveal that MiRF effectively enhances semantic discrimination of fine-grained structures while preserving efficiency, underscoring its strong adaptability to real-world rice phenotyping applications.

TABLE II
EFFECT OF UAV FLIGHT ALTITUDE AND GSD ON RICE PANICLE SEGMENTATION ACCURACY

Height(m)	IoU	F1	*OA*	*GSD(mm/px)*
12	**0.6527**	**0.7859**	**0.9224**	0.45
15	0.6350	0.7743	0.9094	0.56
20	0.6136	0.7573	0.9012	0.68
30	0.5926	0.7432	0.8926	1.03

3) Comparison Study on Various Flying Heights: The UAV flight altitude directly determines the GSD, which significantly affects rice panicle segmentation accuracy. Theoretically, lower altitudes yield smaller GSD and higher spatial resolution, preserving finer panicle textures and edge details. To systematically evaluate this relationship while respecting operational constraints, all source imagery was acquired at 12 m (GSD = 0.45 mm/px), and additional GSD levels corresponding to 15, 20, and 30 m were derived through controlled downsampling to 0.56, 0.68, and 1.03 mm/px, respectively. As shown in Table II, all metrics consistently declined with

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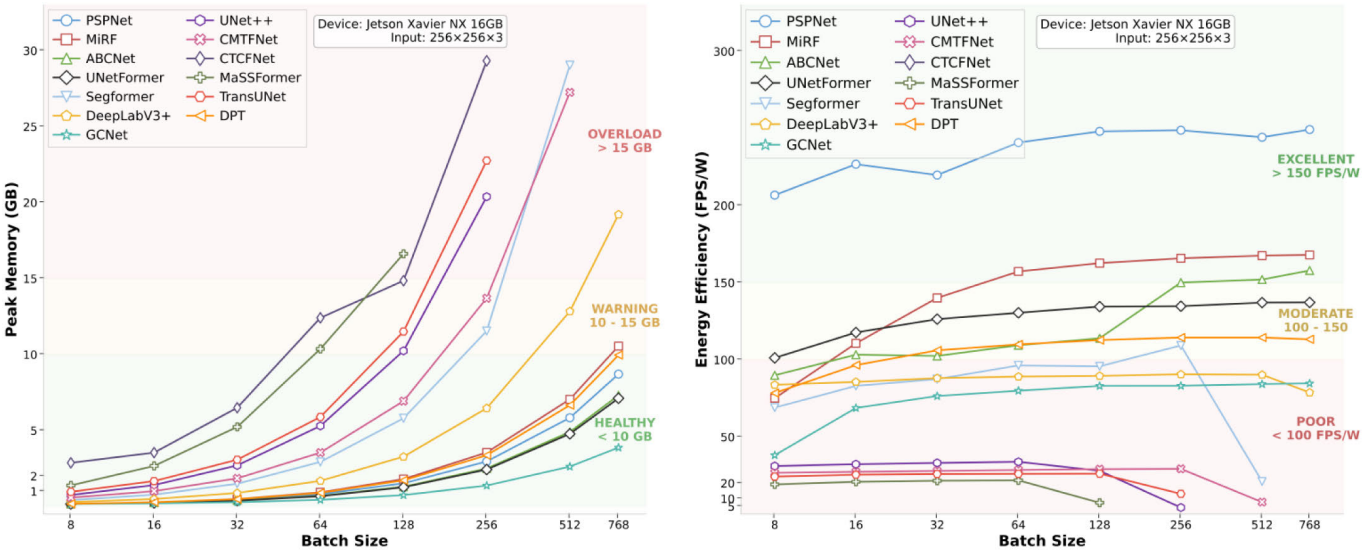


Fig. 9. Hardware-aware simulation of deployment performance on the Jetson Xavier NX edge platform. (Left) Estimated peak memory consumption versus batch size. (Right) Projected energy efficiency (FPS/W) versus batch size.

TABLE III
ABLATION STUDY RESULTS ON UAV_PANICLE AND RICE_SEG_PANICLES DATASETS

Model	*UAV_Panicle*				*Rice_SEG_Panicles*				**Params**
	F1	**IoU**	**OA**	**GFLOPs**	**F1**	**IoU**	**OA**	**GFLOPs**	
Baseline	0.7657	0.6263	0.9133	2.921	0.8494	0.7529	0.9429	11.686	3.13
Baseline+BFP	0.7699	0.6319	0.9147	4.025	0.8606	0.7637	0.9439	16.101	3.57
Baseline+ECFA(CNN)	0.7713	0.6336	0.9146	1.353	0.8615	0.7641	0.9436	5.413	2.789
Baseline+ECFA(DyPConv)	0.7789	0.6433	0.9201	**0.757**	0.8662	0.7734	0.9457	**3.029**	**2.62**
Baseline+ECFA(DyPConv)+FAR	**0.7859**	**0.6527**	**0.9224**	1.048	**0.8780**	**0.7825**	**0.9559**	4.601	2.81

increasing simulated altitude, confirming that higher resolution improves segmentation performance. The highest accuracy was achieved at 12 m (IoU = 0.6527, $F1$ = 0.7859, and OA = 0.9224). Although altitudes below 12 m might theoretically afford marginal gains, 12 m represents the practical optimum that balances phenotypic extraction fidelity against flight safety and the need to protect small microplot breeding materials from rotor downwash damage, particularly for germplasm accessions with unknown mechanical resistance.

4) Simulated Energy Efficiency and Deployment Feasibility on Edge Devices: As illustrated in Fig. 9 and Table I, existing architectures generally face a dilemma to balance accuracy and edge deployment capability. While PSPNet achieves superior energy efficiency, its segmentation accuracy remains constrained. Conversely, SegFormer delivers competitive precision yet suffers from severe memory overhead and computational costs. In contrast, MiRF (MobilenetV2) achieves a better balance among accuracy, energy efficiency, and memory footprint through its extremely lightweight design. It not only demonstrates segmentation precision comparable to complex models in Table I but also exhibits a gracefully ascending memory curve well below the saturation threshold and sustained energy efficiency improvement without performance degradation in Fig. 9. This makes MiRF the sole architecture simultaneously satisfying real-time throughput, memory safety, and high-accuracy requirements for robust edge deployment.

D. Ablation Study

In order to verify the effectiveness of the components of the proposed MiRF, we attempted to add a single component or a combination of components to the baseline for experimental verification. This study utilizes feature pyramid network (FPN) with MobileNetV2 as the baseline. For a more comprehensive comparison, the baseline combined with the bidirectional feature pyramid (BFP) was also included in the ablation experiments to serve as a base reference. Two datasets, including the UAV_Panicle and Rice_SEG_Panicles, were selected to verify the generalization and robustness of each component or combination of components. The reason for this choice is that these two datasets are directly related to panicle segmentation. The numerical results are shown in Table III.

1) Effect Analysis of ECFA: The integration of ECFA demonstrates substantial improvements in both performance and computational efficiency compared to baseline architectures. Baseline + ECFA(CNN) achieves $F1$ scores of 0.7713 and 0.8615 on UAV_Panicle and Rice_SEG datasets, respectively, while reducing model parameters from 3.13 to 2.789 M and dramatically decreasing computational cost from

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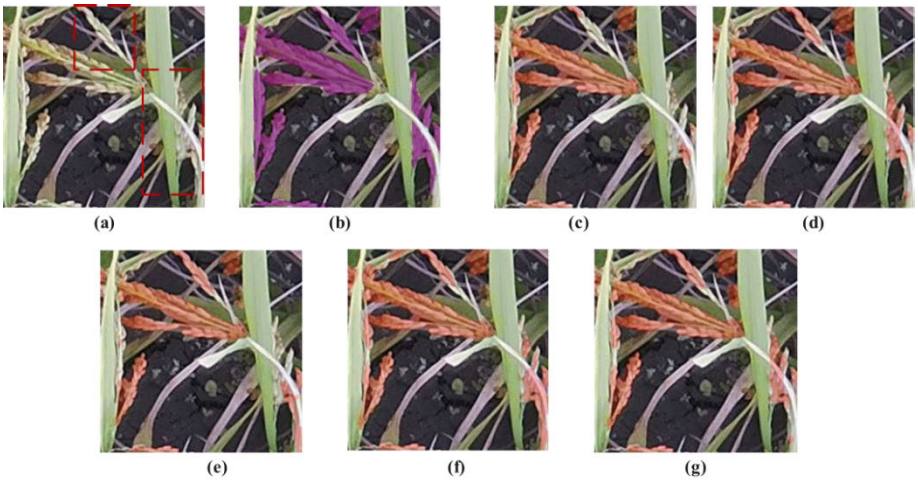


Fig. 10. Ablation study results on semantic segmentation in the UAV_Panicle Dataset. (a) Original. (b) GT. (c) Baseline. (d) Baseline + BFP. (e) Baseline + ECFA(CNN). (f) Baseline + ECFA(DyPConv). (g) Baseline + ECFA(DyPConv) + FAR.

11.686 GFLOPs to 5.413 GFLOPs (53.7% reduction). Compared to Baseline + BFP requiring 3.57 M parameters and 16.101 GFLOPs, the ECFA module is significantly superior in efficiency.

This tradeoff stems from its globally shared fusion weights that enable stable bidirectional feature aggregation without per-layer parameter proliferation and streamlined lateral channel alignment that unifies multiresolution features with minimal overhead. As shown in Figs. 10(e) and 11(e), ECFA achieves nearly identical segmentation performance compared to counterpart architectures while reducing both computational cost and model parameters.

2) Effect Analysis of DyPConv: As shown in Figs. 10(f) and 11(f), the overall structure of panicles is more accurate, and the boundary between panicles and leaves is more clearly depicted. This result indicates that replacing standard convolution with DyPConv yields consistent improvements while further reducing computational burden. As shown in Table III, Baseline + ECFA(DyPConv) achieves $F1$ scores of 0.7789 and 0.8662, with IoU-Panicles increasing from 0.6336 to 0.6433 (+0.97%) on UAV_Panicle and from 0.7641 to 0.7734 (+0.93%) on Rice_SEG. Remarkably, these gains are accompanied by a 44% reduction in computational cost (from 5.413 to 3.029 GFLOPs). This efficiency-accuracy synergy arises from DyPConv's multibranch parallel structure with varying dilation rates that captures multiscale contextual information, the depthwise separable convolution paradigm that substantially reduces redundant computation while preserving expressive capacity, and the preview mechanism that adaptively adjusts receptive fields according to local feature characteristics and reuses previewed features to reduce computational overhead. By adaptively adjusting receptive fields, DyPConv enables the model to better localize and recognize sparse and small panicles, effectively addressing the challenges of small object segmentation in UAV RSI. This adaptive capability is particularly valuable for handling objects with significant scale variations in RSI where rice panicles exhibit diverse sizes depending on growth stages and ground sample distance. The consistent improvements across datasets with different dataset characteristics demonstrate the generalization of DyPConv in different scenarios of RSI semantic segmentation.

3) Effect Analysis of Far: The incorporation of FAR brings performance improvement, which establishes the architecture's superiority with $F1$ scores of 0.7859 and 0.8780, and IoU scores of 0.6527 and 0.7825. As shown in Table III, the improvements are pronounced on UAV_Panicle (+0.70% in $F1$ and +0.94% in IoU), which demonstrates FAR's effectiveness in handling challenging scenarios with cluttered backgrounds and irregular boundaries. While FAR increases computational cost from 3.029 to 4.601 GFLOPs, the final model maintains 61% lower FLOPs than the baseline with superior accuracy. This performance boost stems from the introduced Laplacian high-frequency kernel that explicitly captures high-frequency information, which corresponds to fine-grained textures that are often attenuated in deep networks. Combined with a learnable frequency-dependent weighting mechanism that adaptively enhances important streams and suppresses noise, this design offers a frequency representation that is complementary to spatial convolutions. Together, these components alleviate the inherent limitation of CNNs in modeling sharp transitions, thereby improving performance. As shown in Figs. 10(g) and 11(g), the FAR module refines the boundary information in semantic segmentation, effectively alleviating boundary ambiguity caused by color similarity and interclass similarity. This result demonstrates the effectiveness of a progressive enhancement strategy where ECFA establishes efficient multiscale aggregation, DyPConv enriches contextual representation, and FAR refines boundary precision.

Across different high-frequency kernels, we observe that a moderate level of HF activation is beneficial for the three-branch architecture. The HF branch adds discriminative cues along edges and fine textures, yielding clearer boundaries and more stable predictions for small targets. However, when the HF kernel becomes overly "aggressive," the HF branch

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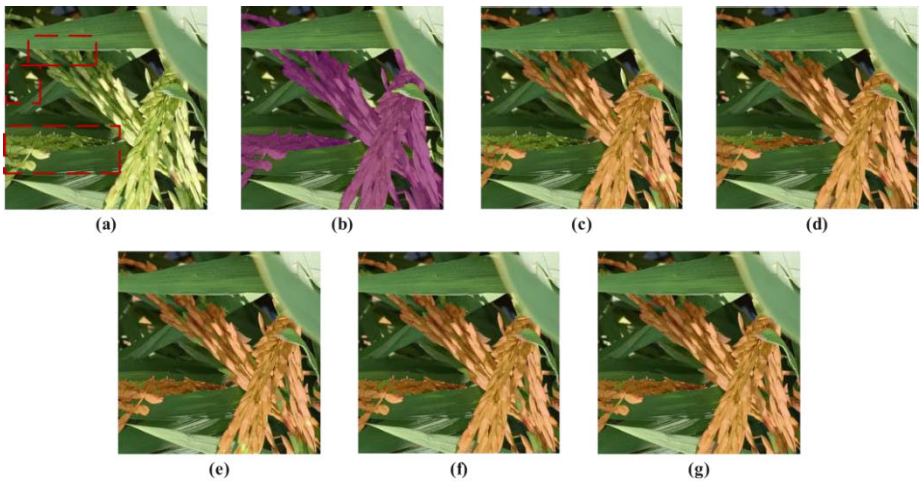


Fig. 11. Ablation study results on semantic segmentation in the Rice_SEG_Panicles dataset. (a) Original. (b) GT. (c) Baseline. (d) Baseline + BFP. (e) Baseline + ECFA(CNN). (f) Baseline + ECFA(DyPConv). (g) Baseline + ECFA(DyPConv) + FAR.

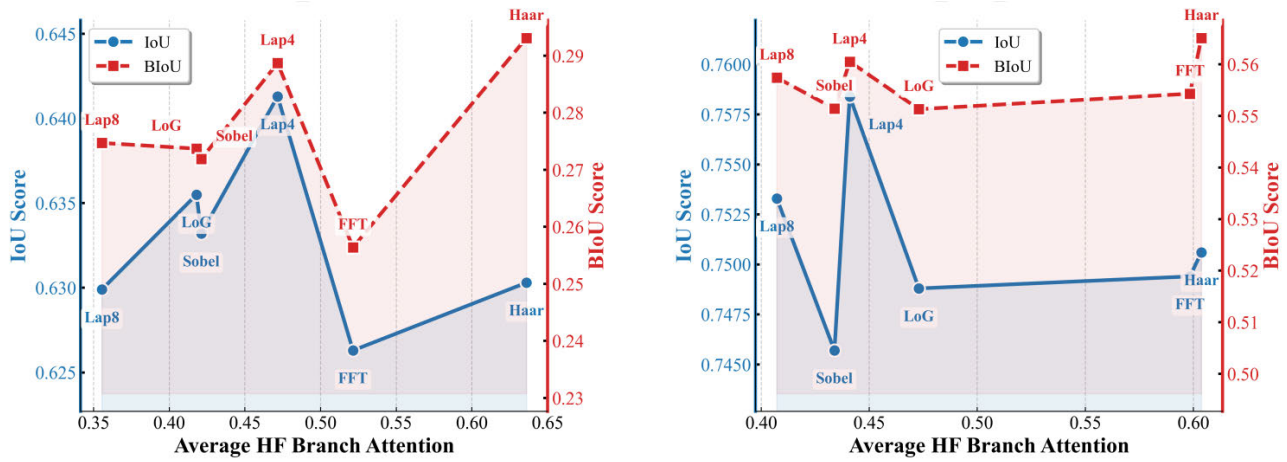


Fig. 12. Relationship between average HF-branch attention and segmentation accuracy for different high-frequency kernels in UAV_Panicle (left) and Rice_SEG_Panicles (right).

TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT HIGH-FREQUENCY KERNELS IN FAR ON UAV_PANICLE AND RICE_SEG_PANICLES

Model	UAV_Panicle				Rice_SEG_Panicles			
	F1	IoU	BIoU	OA	F1	IoU	BIoU	OA
Lap4(4-neighborhood Laplacian)	**0.7814**	**0.6413**	0.2887	**0.9154**	**0.8597**	**0.7584**	0.5605	0.9490
Lap8(8-neighborhood Laplacian)	0.7729	0.6299	0.2747	0.9085	0.8593	0.7533	0.5574	**0.9494**
LoG(Laplacian of Gaussian)	0.7771	0.6355	0.2737	0.9107	0.8563	0.7488	0.5513	0.9474
Sobel	0.7754	0.6332	0.2719	0.9116	0.8543	0.7457	0.5514	0.9481
FFT	0.7702	0.6263	0.2564	0.9081	0.8568	0.7494	0.5543	0.9486
Haar	0.7732	0.6303	**0.2931**	0.9103	0.8575	0.7506	**0.5651**	0.9486

produces excessively high-confidence responses on many true and pseudoedges, simultaneously amplifying object boundaries, noise, repetitive textures, and background details, which interferes with the discriminative capability of the main spatial branches. As shown in Fig. 12, the average HF attention curves and HF attention weights indicate that once the HF branch dominates, the energy distribution among branches becomes unbalanced and degrades robustness. In this case, attention is overconcentrated on boundaries, causing inconsistencies inside objects and confusion between categories. Quantitative results in Table IV also confirm that FAR benefits from an appropriate level of HF enhancement rather than simply pursuing stronger high-frequency responses.

4) Effectiveness of the Preview-Based Dynamic Branch Selection: Fig. 13 illustrates the relationship between DyPConv activations and image structural features,

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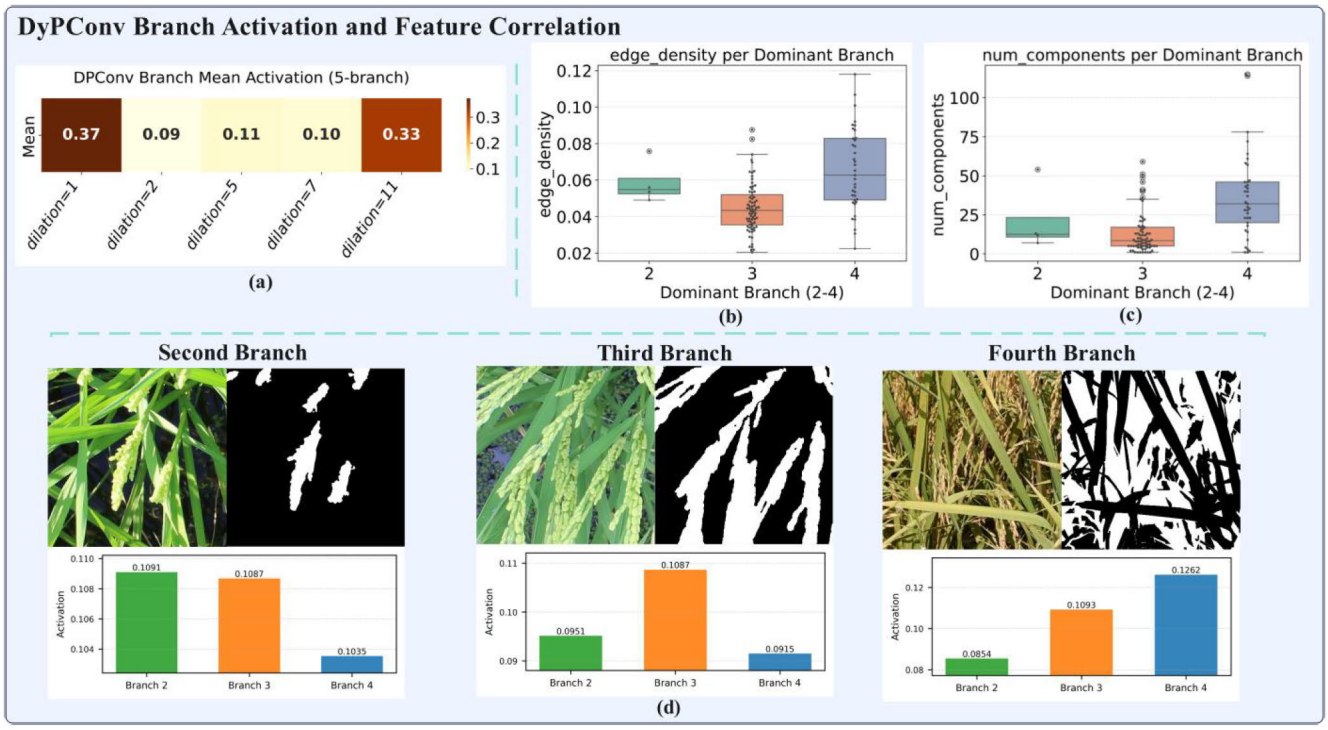


Fig. 13. DyPConv branch activation, feature correlation, and branch selection in sparse-dense scenarios. (a) Mean Branch Activation. (b) Edge Density Distribution. (c) Connected Component Distribution. (d) Texture Adaptive Branch Selection.

highlighting its interpretability. In Fig. 13(a), the heatmap of average activations across five dilation branches shows that the branches with dilation = 1 and 11 possess the highest weights, indicating that DyPConv effectively integrates local details with global contextual information to achieve complementary feature representation. Fig. 13(b) and (c) depicts the distributions of edge density and connected components for different dominant branches, suggesting that branches with larger dilation rates tend to capture regions characterized by sparse textures and complex structures. Notably, DyPConv's dynamic branch selection follows a TOP-3 selection mechanism, where the third branch adaptively switches among dilations = 2, 5, and 7 depending on task requirements: when the input region is relatively sparse, dilation = 2 is activated to enhance fine details; for moderately sparse or structurally uniform areas, dilation = 5 is chosen to strengthen global representation while preserving details; and for dense and intricate regions, dilation = 7 is activated to reinforce global structural coherence. Fig. 13(d) further exemplifies this behavior with representative samples from the second to fourth branches (dilations = 2, 5, and 7): the second branch focuses on fine and well-defined textures, the third branch responds to moderately complex structures, and the fourth branch emphasizes regions with complex texture. Overall, DyPConv demonstrates an adaptive multiscale selection mechanism in semantic segmentation, dynamically balancing local context and global structure according to the spatial density of the input image.

The quantitative results in Table V further demonstrate the influence of the number of selected branches (k) on segmentation performance and computational efficiency. As the value of k increases, the model's IoU increases from 0.5259 at Top-1 to 0.7825 at Top-3, indicating that integrating multiple receptive fields effectively enhances the model's ability to capture both local and global context information. However, when more branches are included (Top-4 and Top-5), performance begins to decline despite higher computational cost, suggesting that excessive receptive field diversity introduces redundancy or feature noise, which weakens feature learning and decreases segmentation accuracy. Conversely, selecting too few branches limits the model's ability to generalize across varying texture and structure scales, leading to incomplete contextual perception. Therefore, choosing an appropriate number of active branches (Top-3 in this case) strikes a balance between representational richness and noise suppression, enabling DyPConv to maintain efficient computation while achieving optimal segmentation performance.

TABLE V
ABLATION STUDY OF DYPCONV BRANCH NUMBER (TOP-K) IN MIRF MODEL ON VALIDATION SET

branch_num(Top-K)	IoU	GFLOPs
1	0.5259	3.896
2	0.7425	4.249
3	0.7825	4.601
4	0.7742	4.953
5	0.7622	5.306

E. Application of the Proposed MiRF Method for Rice Yield Prediction

A segmentation-based yield prediction framework was constructed to validate the practical applicability of the proposed

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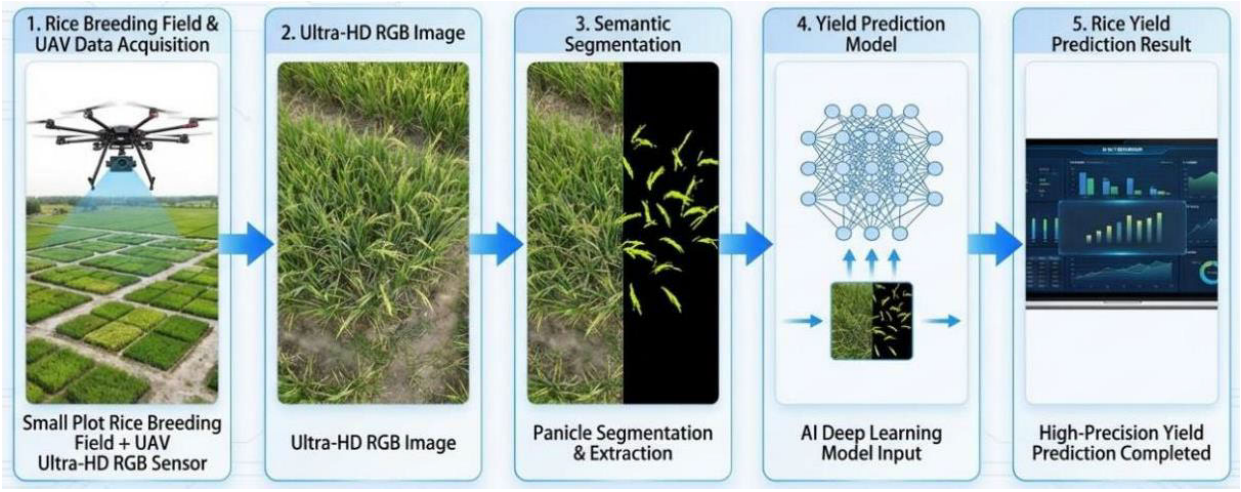


Fig. 14. Process of segmentation-based yield prediction in rice breeding.

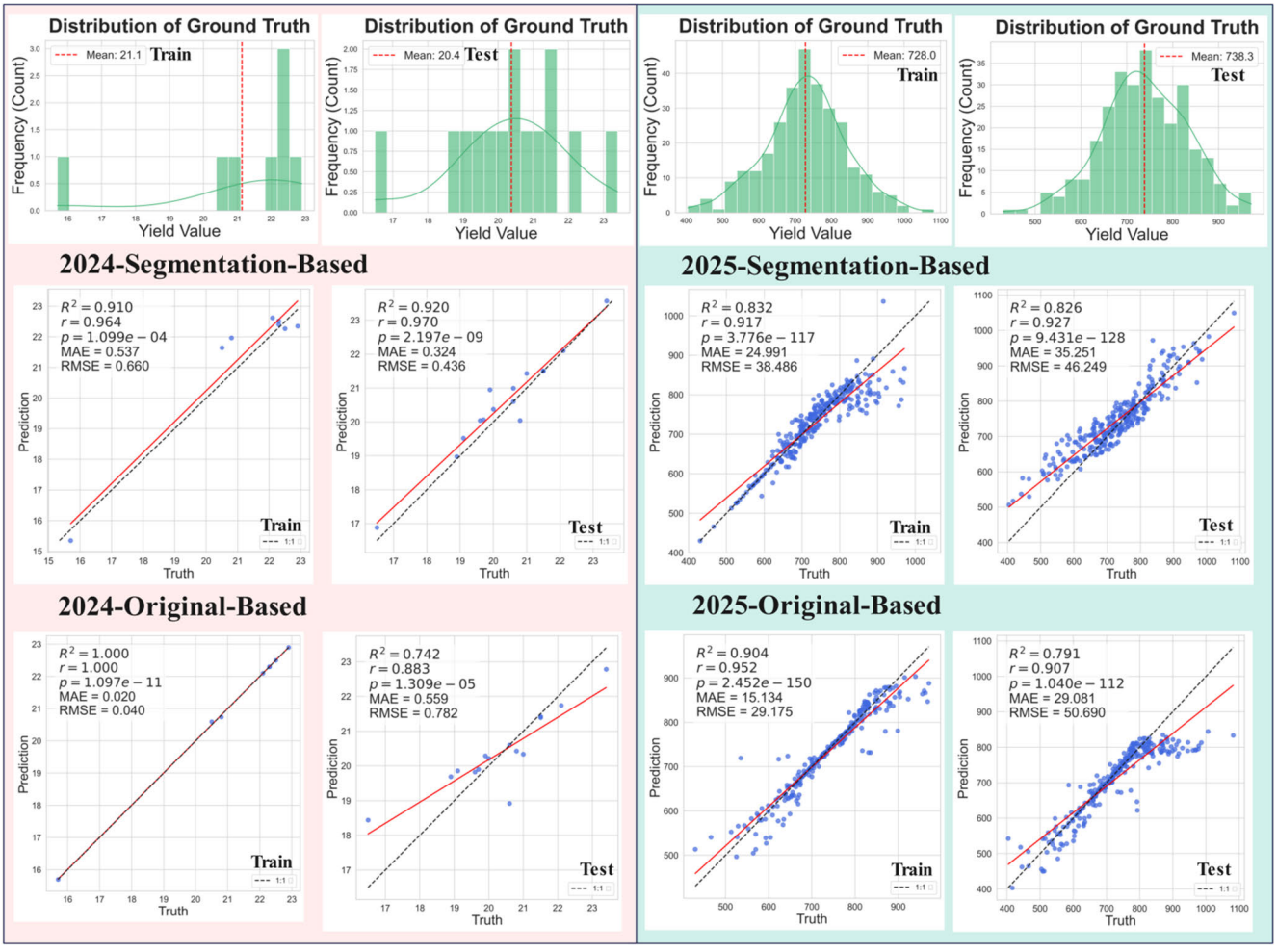


Fig. 15. Comparison of cross-year rice yield prediction results under panicle segmentation- and original-based inputs.

method, as illustrated in Fig. 14. The workflow starts by applying MiRF-generated binary masks to the original UAV imagery to extract panicle ROI, effectively eliminating background interference such as soil and shadowed leaves. The segmented RGB images serve as inputs to a customized lightweight residual regression network, the detailed architecture of which is elaborated on in Section III-D.

The quantitative evaluation demonstrates the model's robustness across different planting scales and seasons, as illustrated in Fig. 15. For the 2024 experiment conducted

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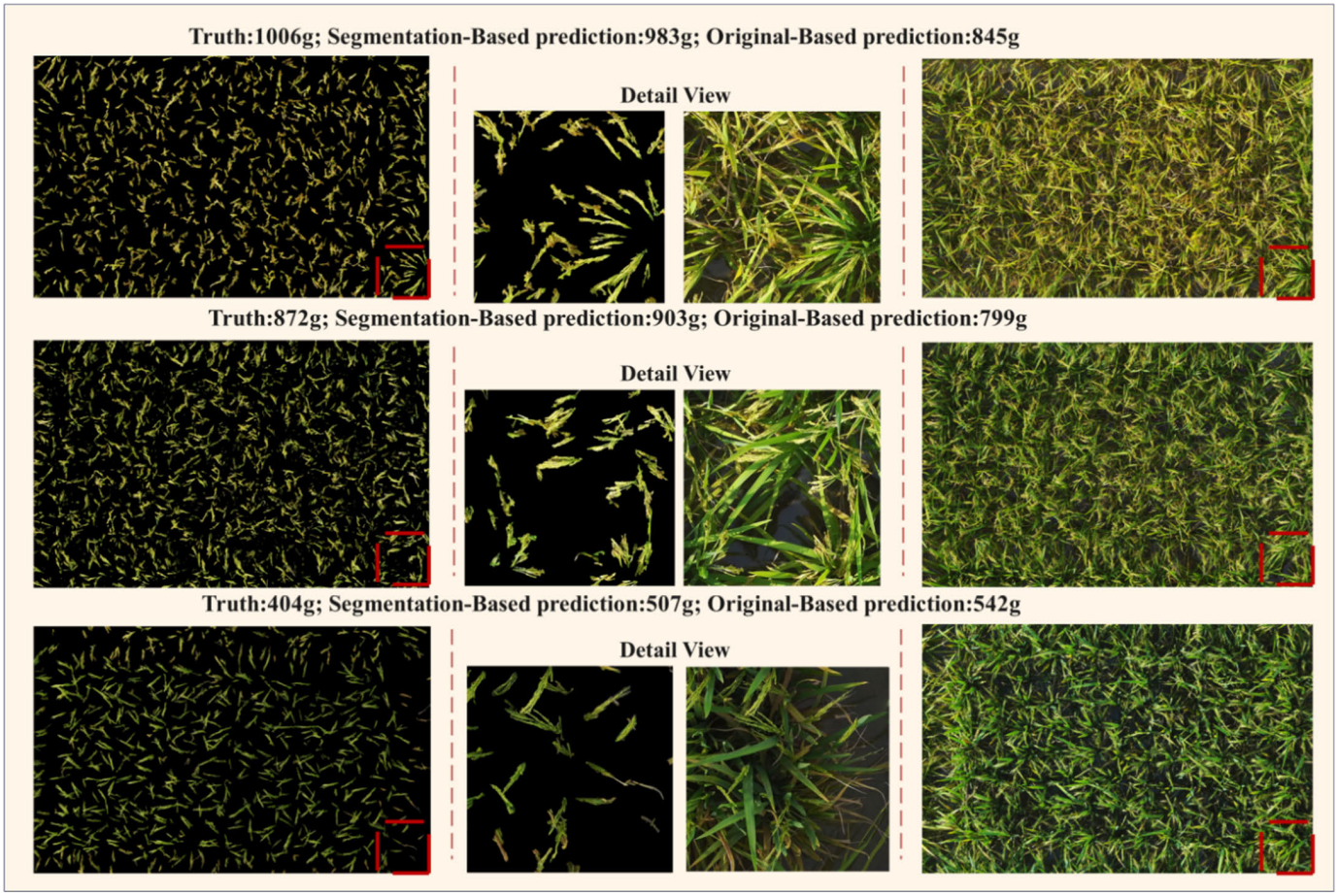


Fig. 16. Comparison of input images for yield prediction. The left column displays the segmented rice panicle results used as model input, while the right column shows the original raw images used as input. The center columns present detailed views of the regions marked by red boxes.

on large-scale field plots, the segmentation-based model achieved exceptional accuracy, with an R^2 of 0.910 (RMSE = 0.660 kg) in training and an R^2 of 0.920 (RMSE = 0.436 kg) in testing. To assess adaptability, the framework was applied to the 2025 dataset involving smaller experimental plots with significantly increased variety diversity. Despite the variation in spatial scale and measurement units, the model maintained consistent predictive performance (training: $R^2 = 0.832$, RMSE = 38.486 g; testing: $R^2 = 0.826$, and RMSE = 46.249 g).

To verify the necessity of the segmentation module, comparative experiments using original RGB images were conducted. In the 2024 dataset, the original-based model exhibited severe overfitting, achieving a perfect R^2 of 1.000 in training but dropping sharply to 0.742 in testing. This suggests that with limited sample size, the deep network tended to memorize background noise rather than learning generalizable yield traits. In contrast, the 2025 original-based model showed improved generalization (testing $R^2 = 0.791$) due to the larger dataset. However, a critical ceiling effect was observed in the 2025 original-based testing phase: for high-yield samples, predictions plateaued and converged toward the dataset mean. This phenomenon is attributed to feature saturation and data distribution imbalance.

This segmentation-based yield prediction can be explained by crop physiological principles. Projected panicle area serves as a physiological proxy for sink capacity through the source–sink relationship, where greater panicle coverage correlates with higher spikelet numbers and enhanced light interception efficiency [48]. The ultrahigh resolution (0.45 mm/px) enables pixel area to serve as a meaningful surrogate for panicle volume via allometric scaling [49]. Meanwhile, the segmentation masks implicitly encode panicle number per unit area, and deep features extracted from RGB imagery implicitly characterize grain quality indicators, including ripeness and plumpness [9]. As shown in Fig. 16, one of the primary reasons for the ceiling effect is that high-yield rice panicles tend to be fuller and denser, causing them to droop more easily. These drooping panicles are readily obscured by leaves, making it challenging for the model to effectively identify critical characteristics of the rice, particularly under complex backgrounds. Since the aerial images captured by drones rely on orthogonal views, using complete original images introduces substantial noise. Furthermore, the drooping of the panicles reduces the visibility of critical information, amplifying this noise and raising the difficulty of accurately capturing relevant features, especially in the presence of abundant yellow leaves. Crucially, models based on segmented inputs effectively filter out the majority of this noise during the segmentation process. While this procedure may inevitably discard a fraction of yield-correlated biological details in the context of high-yield original images, the interference introduced by yellow leaves

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and drooping panicles far outweighs the value of the effective features sacrificed during segmentation. Consequently, despite the fragmentation, the number of panicles per unit area remains relatively stable. This contributes to the robustness of yield predictions based on segmented images, allowing the model to better account for the essential characteristics of high-yield varieties and mitigating the ceiling effect typically associated with original images.

V. Discussion

A. Limitations of the Static Laplacian Kernel

The FAR module employs a statically predefined Laplacian-4 kernel for high-frequency extraction, reflecting a tradeoff to avoid the computational overhead of adaptive kernel learning, which is critical for ensuring real-time inference on resource-constrained edge devices. While this design achieves optimal performance under the experimental conditions of this research, we acknowledge that its performance under substantially different lighting environments or varying sensor characteristics remains to be further explored, and adaptive or learnable high-pass filters represent a promising direction for future multisite deployment.

B. Segmentation Error Propagation and Biological Transfer Chain: Future Directions

Two important research directions remain valuable for strengthening the research of this article. First, a formal sensitivity analysis quantifying how segmentation errors propagate into yield prediction accuracy would measure the influence of segmentation quality on the downstream yield prediction. Such analysis requires pixel-level panicle mask annotations on the yield prediction datasets, which are not currently available and would necessitate extensive annotation campaigns. Second, direct validation of the biological transfer chain, correlating measured panicle count, independently measured coverage area, and yield on our experimental dataset, would provide stronger mechanistic grounding. However, the current yield datasets do not cover the plot-level metrics of panicle counts and independent coverage measurements, and the ground investigation for these parameters requires a large amount of manual labor. Therefore, all these works will be organized as our future work in subsequent growing seasons.

In addition, the strong yield prediction performance ($R^2 > 0.826$) reflects generalization for the tested varieties. However, the field tests mainly cover the advanced-generation cultivars with relatively close agronomic traits. For early-generation germplasm resources or highly heterogeneous genetic populations, yield-related traits show greater variance. The generalization of our pipeline to these scenarios remains to be further evaluated.

VI. Conclusion

This study presents MiRF, a lightweight yet discriminative framework for UAV-based rice panicle segmentation and segmentation-guided yield prediction under complex breeding-field conditions. The motivation of this work stems from three persistent bottlenecks in high-throughput rice phenotyping: scale heterogeneity among genotypes, severe foreground/background ambiguity caused by canopy occlusion, and the computational burden imposed by UHR orthomosaics. To address these challenges, MiRF integrates three complementary designs. DyPConv introduces an input-adaptive receptive-field selection mechanism that activates only the most informative dilation branches, thereby improving scale-aware representation while avoiding redundant computation. FAR explicitly enhances high-frequency and boundary-sensitive responses through a learnable fusion of spatial, Laplacian, and identity streams, strengthening the discrimination between panicles and visually similar senescent leaves. ECFA further enables efficient bidirectional cross-scale aggregation with shared fusion parameters and inference-oriented reparameterization, improving the balance between accuracy and deployability.

Extensive experiments demonstrate that MiRF consistently outperforms representative CNN-, Transformer-, and hybrid-based segmentation models on two rice panicle benchmarks. The observed gains in panicle IoU and $F1$-score confirm that accurate organ-level segmentation requires not only global semantic context but also adaptive local receptive fields and frequency-aware boundary refinement. The lightweight MobileNetV2 variant further shows favorable efficiency, memory behavior, and simulated edge-deployment potential, indicating that MiRF is suitable for practical real-time image processing on edge devices. Beyond segmentation, the integration of MiRF-derived panicle masks with a lightweight regression model achieves robust yield prediction across 610 rice varieties spanning two cropping seasons, suggesting that pixel-level panicle morphology can provide biologically meaningful cues for large-scale breeding evaluation.

Nevertheless, several limitations still remain. DyPConv may introduce additional intermediate memory consumption, FAR currently relies on fixed Laplacian priors, and RGB-based yield estimation is still constrained under heavily occluded conditions. Future work will explore more memory-efficient dynamic selection strategies, adaptive frequency operators, and multimodal fusion with multispectral, thermal, or LiDAR data. These extensions may further improve cross-domain robustness and support a unified, field-deployable phenotyping system for rice breeding and precision agriculture.

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Zibin Zhou received the M.Sc. degree in agricultural engineering and information technology from the College of Engineering, South China Agricultural University, Guangzhou, China, in 2023. He is currently pursuing the Ph.D. degree in Agricultural Engineering with the College of Mechanical and Electrical Engineering, Fujian Agriculture and Forestry University, Fuzhou, China.

His research interests include UAV remote sensing, image processing, and UAV-based crop phenotyping.



Xuan Wei received the B.Sc. degree in food quality and safety engineering from Jiangnan University, Wuxi, China, in 2009, and the Ph.D. degree in biological systems engineering from Zhejiang University, Hangzhou, China, in 2014.

From 2013 to 2014, she was a Joint Ph.D. Researcher with the University of Arizona, Tucson, AZ, USA. She is currently an Associate Professor with the College of Mechanical and Electrical Engineering, Fujian Agriculture and Forestry University, Fuzhou, Fujian, China. Her research interests include rapid detection methods, flexible and wearable sensors, and agricultural sensors.



Huasheng Huang received the M.Sc. degree in pattern recognition and the Ph.D. degree in agricultural electrification from South China Agricultural University, Guangzhou, China, in 2013 and 2019, respectively.

He is currently an Associate Professor with the School of Artificial Intelligence, Guangdong Polytechnic Normal University, Guangzhou. His research interests include image processing, UAV remote sensing, and intelligent agriculture.



Tao Guo received the M.Sc. and Ph.D. degrees in agronomy from the College of Agriculture, South China Agricultural University, Guangzhou, China, in 2004 and 2012, respectively.

He is currently a Professor with the College of Agriculture, South China Agricultural University. His research expertise lies in developing innovative rice breeding technologies, including space biology studies, genome editing, and the creation of advanced haploid induction lines, aimed at accelerating the breeding of rice varieties with superior traits.



Jiajun Zhuang received the B.Sc. and M.Sc. degrees in measurement and control technology from Guangdong University of Technology, Guangzhou, China, in 2007 and 2010, respectively, and the Ph.D. degree in computer science from South China University of Technology, Guangzhou, in 2013.

He is currently an Associate Professor with the College of Mathematics and Data Science, Zhongkai University of Agriculture and Engineering, Guangzhou. His research interests include computer vision, machine learning, and agricultural engineering.



Zhiping Tan received the B.Sc. degree in electronic engineering from Xiangtan University, Xiangtan, China, in 2015, and the Ph.D. degree from the College of Mathematics and Informatics, South China Agricultural University, Guangzhou, China, in 2021.

He is currently an Associate Professor with the Academy of Interdisciplinary Studies, Guangdong Polytechnic Normal University, Guangzhou. His research interests include evolutionary optimization, deep learning, and machine vision.



Jinchang Ren (Senior Member, IEEE) received the B.Eng. degree in computer software, the M.Eng. degree in image processing, and the D.Eng. degree in computer vision from Northwestern Polytechnical University, Xi'an, China, in 1992, 1997, and 2000, respectively, and the Ph.D. degree in electronic imaging and media communication from the University of Bradford, Bradford, U.K., in 2009.

He is currently a Professor with the School of Artificial Intelligence, Guangdong Polytechnic Normal University, Guangzhou, China, with more than 300 scientific articles. His research interests include image processing, machine learning, hyperspectral imaging, remote sensing, and big data analytics.



Yu Tang (Member, IEEE) received the B.Sc. degree in electrical engineering from the Civil Aviation University of China, Tianjin, China, in 2005, and the M.S. degree in optical engineering and the Ph.D. degree in microelectronics and solid-state electronics from South China University of Technology, Guangzhou, China, in 2008 and 2013, respectively.

From 2013 to 2015, he worked as a Post-Doctoral Researcher at South China University of Technology. He is currently a Professor with the Academy of Interdisciplinary Studies, Guangdong Polytechnic Normal University, Guangzhou, and the Vice President with Lingnan Normal University, Zhanjiang, China. His research interests include artificial intelligence, image/video processing, and UAV remote sensing.

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